

NGUYEN THANH PHU

Phu Thuan Ward, HCMC | (+84) 845939722 | msnp@outlook.com.vn | github.com/ngxx-fus



SUMMARY

Embedded engineer passionate about low-level programming and hardware-software integration..

EDUCATION

Ho Chi Minh City University of Technology and Engineering
GPA: 3.28/4

Expected Graduation 2026
Major: Computer Engineering Technology

WORK EXPERIENCE

Ban Vien Corporation, Ho Chi Minh City, Viet Nam
Embedded Software Engineer

May, 2026 – Present

Overall:

- Maintained drivers within Renesas RA-FSP, focusing on the RA8 series (Cortex-M85) architecture.
- Resolved customer-reported bugs regarding e2 studio configuration constraints by debugging and updating Jinja templates for accurate FSP code generation.

Achievement gained:

- Completed maintained test-source for DCache-related tickets, currently pending merge for future FSP releases.
- Successfully resolved and merged customer-reported E2 studio configuration issues, including major-priority tickets.

Bosch Vietnam Co., Ltd., Ho Chi Minh City, Viet Nam
System Test Engineer Intern

Oct, 2025 – Apr, 2026

Overall:

- Implemented automated CAPL scripts to validate CAN routing integrity and performance against DBC/ARXML specifications, optimizing testing efficiency through batch-mode processing.
- Performed high-load stress testing to identify data loss thresholds and developed a reusable CAPL library for comprehensive FIFO/MPFO validation.

Achievement gained:

- Reduced test execution time by 15% by transitioning from 1-by-1 message validation to a high-performance batch-checking logic.
- Successfully deployed a reusable validation framework to secondary projects, significantly accelerating the pre-implementation phase.

PROJECTS EXPERIENCE

Zonal ECU Implementation over Ethernet Backbone, Independent Project

2026

Overall:

- Designed and implemented a modular Zonal ECU (ZECU) firmware on ESP32 (FreeRTOS/ESP-IDF), interfacing with an external W5500 Ethernet controller via 20 MHz SPI.
- Developed a SOME/IP-inspired SOA stack over UDP—featuring service discovery, periodic heartbeat, event-driven motor control, and handshake authentication—along with a PySide6 dashboard for real-time monitoring and control.
- Link: github.com/ngxx-fus/ZonalArchECU_W_SomeIP

Achievement gained:

- Achieved an average end-to-end response latency of 54.78 ms for closed-loop motor control and state feedback over the Ethernet backbone.
- Designed a fixed-size 612-byte PDU structure with End-to-End (E2E) protection (Fast-path Checksum + CRC-32), eliminating IP fragmentation and optimizing socket memory buffer allocation.
- Successfully built a non-blocking, interrupt-driven network pipeline using a FreeRTOS Sleep-Wakeup mechanism and thread-safe queues, preventing CPU starvation and ensuring reliable real-time sensor processing.

Overall:

- Developed a lightweight clipboard monitoring utility in C/X11 on Void Linux to capture and store copied text and images. Implemented IPC via signal-based interaction with Rofi for search and history retrieval.
- Link: github.com/ngxx-fus/xClipboardCapture

Achievement gained:

- Achieved a minimal memory footprint (under 5MB RAM usage), ensuring zero impact on system performance during continuous background monitoring.

Smart Decorative Screen, Independent Project

2025

Overall:

- Built an ESP32-based device with a 2.2-inch TFT LCD for image display, environmental monitoring, and user interaction.
- Implemented SD card image rendering, temperature and humidity monitoring, lighting control, and a 5-button navigation interface.
- GitHub: github.com/ngxx-fus/SmartDecorativeScreen

Achievement gained:

- Successfully integrated multiple peripherals (TFT LCD, SD Card, DHT Sensor) into a single, compact ESP32-based system with zero hardware-software conflicts.

Designed a responsive 5-button menu system, providing an intuitive user experience for device navigation and lighting control.

TECHNICAL SKILLS

- Programming Languages: C, C++, Python, Bash, CAPL Script.
- Working Environments: Linux (Void Linux, Ubuntu)
- Tools & Platforms: Vector CANoe, Git, GitHub, Qt, Valgrind, Cadence Virtuoso, Altium Designer.
- Protocols: CAN/CAN FD, LIN, UART, SPI, I2C.
- MCU & HW: MCU (ESP32, AT89C52), RPi, FreeRTOS, digital logic design, OOP, DSA.
- Others: Fundamental networking knowledge (NAT, PAT, routing).

CERTIFICATIONS & AWARDS

- Engineering Projects in Community Service (Season 7). 2023
- Samsung Innovation Campus - Internet of Things course.

VOLUNTEERING

- Blood Donation at Cho Ray Hospital 2025

EOF